Haibo Li

School of Mathematics and Statistics, Huazhong University of Science and Technology
Luoyu Road 1037, Wuhan, Hubei 430074, China
Email: haibolee1729@gmail.com, haiboli@hust.edu.cn
Homepage: https://haiboli99.github.io/

Current Position

Associate Professor, School of Mathematics and Statistics, Huazhong University of Science and Technology, Dec. 2025 – now

Research Interest

My research aims to build rigorous mathematical theory and design efficient algorithms for real-world challenges in science, engineering, and beyond. The topics mainly include:

- Inverse Problems
- Numerical Linear Algebra
- Scientific Machine Learning

Work Experience

Sept. 2023 – Oct. 2025	• Research Fellow, School of Mathematics and Statistics, The University of Melbourne, (supervisor: Hailong Guo)
Aug. 2024 – Oct. 2024	• Visiting Scholar, Department of Mathematics, Johns Hopkins University (host: Fei Lu)
Aug. 2021 – Sept. 2023	• Researcher, Computing System Optimization Lab, Huawei Technologies Co., Ltd.

Education

Ph.D. in Computational Mathematics	• Department of Mathematical Sciences. Tsinghua University, China, 2015.9 – 2021.6
B.S. in Mathematics	• Taishan College (talent training project), 2012.9 – 2015.6; School of Mathematics, 2011.9 – 2012.9. Shandong University, China

Papers

Publications:

- **Haibo Li**, Jinchao Feng, Fei Lu. *Scalable iterative data-adaptive RKHS regularization*, SIAM Journal on Scientific Computing, accepted, 2026.
- **Haibo Li**. *Subspace projection regularization for large-scale Bayesian linear inverse problems*, Applied Numerical Mathematics, 225, 55–75, 2026
- **Haibo Li**. *Generalizing the SVD of a matrix under non-standard inner product and its applications to linear ill-posed problems*, Applied Mathematics and Computation, 508(1), 1–23, 2026.
- **Haibo Li**. *Krylov iterative methods for linear least squares problems with linear equality constraints*, Numerical Algorithms, Published online 09 August 2025.
- **Haibo Li**. *Projected Newton method for large-scale Bayesian linear inverse problems*, SIAM Journal on Optimization, 35(3), 1439–1468, 2025.
- **Haibo Li**. *A new interpretation of the weighted pseudoinverse and its applications*, SIAM Journal on Matrix Analysis and Applications, 46 (2), 934-956, 2025.
- **Haibo Li**. *Characterizing GSVD by singular value expansion of linear operators and its computation*, SIAM Journal on Matrix Analysis and Applications 46 (1), 439-465, 2025.

Papers (continued)

- Rongrong Liu, Zhuoqiang Guo, Qiuchen Sha, Tong Zhao, **Haibo Li**, Wei Hu, Lijun Liu, Guangming Tan, Weile Jia. *Large Scale Finite-Temperature Real-time Time Dependent Density Functional Theory Calculation with Hybrid Functional on ARM and GPU Systems*, International Parallel and Distributed Processing Symposium (IPDPS 2025)
- **Haibo Li**, Guangming Tan, Tong Zhao. *Backward error analysis of the Lanczos bidiagonalization with reorthogonalization*, Journal of Computational and Applied Mathematics, 116414, 2025.
- **Haibo Li**. *A preconditioned Krylov subspace method for linear inverse problems with general-form Tikhonov regularization*, SIAM Journal on Scientific Computing, 46 (4), A2607-A2633, 2024.
- **Haibo Li**. *The joint bidiagonalization of a matrix pair with inaccurate inner iterations*, SIAM Journal on Matrix Analysis and Applications, 45 (1), 232–259, 2024.
- **Haibo Li**. *Double precision is not necessary for LSQR for solving discrete linear ill-posed problems*, Journal of Scientific Computing, 98 (3), 55, 2024.
- Yujin Yan, **Haibo Li**, Tong Zhao, Lin-Wang Wang, Lin Shi, Tao Liu, Guangming Tan, Weile Jia, Ninghui Sun. *10-million atoms simulation of first-principle package LS3DF on Sugon supercomputer*, Journal of Computer Science and Technology, 39 (1), 45-62, 2024.
- Zhongxiao Jia, **Haibo Li**. *The joint bidiagonalization method for large GSVD computations in finite precision*, SIAM Journal on Matrix Analysis and Applications, 44 (1), 382–407, 2023.
- Zhongxiao Jia, **Haibo Li**. *The joint bidiagonalization process with partial reorthogonalization*, Numerical Algorithms (88), 965–992, 2021.

Preprints:

- **Haibo Li**, Fei Lu. *Automatic kernel regression and regularization for learning convolution kernels*, 2025, submitted.
- Hailong Guo, **Haibo Li**. *Derivative estimation with RKHS for learning dynamics from time-series data*, 2025, submitted.
- **Haibo Li**, Xingxing Wu, Liping Liu, Long Wang, Lin-Wang Wang, Guangming Tan, Weile Jia, *ALKPU: an active learning method for the DeePMD model with Kalman filter*, 2024, submitted.

Presentations

- Invited talk: *Scalable iterative data-adaptive RKHS regularization for linear inverse problems*, Mini-symposium at The 27th Conference of the International Linear Algebra Society (ILAS 2026), Virginia Tech, Blacksburg, May 18–22, 2026.
- Invited talk: *Automatic reproducing kernel and iterative regularization for learning convolution kernels*, School of Mathematics, Sun Yat-Sen University, Jan. 23, 2026.
- Invited talk: *Iterative data adaptive RKHS regularization for learning convolution kernels*, Seminar talk for Computational Neuroscience Lab at Monash University, Oct. 15, 2025.
- Invited talk: *Projected Newton method for large-scale Bayesian linear inverse problems*, Applied Math Seminar at The University of Melbourne, Sept. 25, 2025.
- Invited talk: *Scalable iterative data-adaptive RKHS regularization*, The Eighth Young Scholar Symposium, East Asia Section of Inverse Problems International Association, Chansha, Nov. 17, 2024.
- Invited talk: *Projected Newton method for large-scale Bayesian linear inverse problems*, School of Sciences, Great Bay University, Nov. 6, 2024.
- Invited talk: *Projected Newton method for large-scale Bayesian linear inverse problems*, School of Mathematics, Hunan University, Nov. 5, 2024.
- Talk: *Scalable iterative data-adaptive RKHS regularization*, Applied Math Seminar at Lehigh University, Oct. 24, 2024.
- Talk: *A preconditioned Krylov subspace method for regularizing linear inverse problems*, Data Science Seminar at Johns Hopkins University, Oct. 02, 2024.

Presentations (continued)

- Talk: *Subspace projection regularization for high-dimensional Bayesian inverse problems*, MATRIX research program: Multivariate Dependence Modeling: Theory and Applications, Melbourne, Jul. 30, 2024.
- Talk: *Scalable iterative data-adaptive RKHS regularization*, International Conference on Scientific Computation and Differential Equations (SciCADE 2024), Singapore, Jul. 16, 2024.
- Talk: *Subspace Projection Regularization: preconditioned Golub-Kahan bidiagonalization methods for regularizing linear inverse problems*, The 67th Annual Meeting of the Australian Mathematical Society (AustMS 2023), Brisbane, Dec. 2023.
- Talk: *A mixed precision variant of LSQR for solving discrete linear ill-posed problems*, Research Center for Mathematics and Interdisciplinary Sciences, Shandong University, Mar. 2023.
- Talk: *A Kalman filter based optimizer for training the neural network force field with first-principles accuracy*, Forum for High Performance Computing and Industrial Material Simulations, CCF HPC China, Dec. 2022.
- Talk: *Introduction to iterative algorithms for solving large scale ill-posed problems*, Forum in Mathematics and Interdisciplinary Sciences, Research Center for Mathematics and Interdisciplinary Sciences, Shandong University, Nov. 2019.
- Technical report: *Preconditioned MINRES algorithm: basic theory and implementation*, CAEP Software Center for High Performance Numerical Simulation, Sep. 2019

Academic service

Reviewer of journals/conferences:

- SIAM Journal on Matrix Analysis and Applications
- BIT Numerical Mathematics
- Numerical Algorithms
- ZAMM - Journal of Applied Mathematics and Mechanics
- Journal of Computational and Applied Mathematics
- Applied Mathematics and Computation
- The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC22), 2022

Reviewer of grants:

- International reviewer for Czech Science Foundation (Czech Republic).

Reviewer for Mathematical Reviews (No. 161348)

Teaching Experience

- Mathematical Analysis (problem sessions), Huazhong University of Science and Technology, 2026.3–
- Stochastic Mathematical Methods (for undergraduates), teaching assistant, Tsinghua University. 2015.9-2016.1
- Stochastic Mathematical Methods (for undergraduates), teaching assistant, Tsinghua University. 2016.2-2016.6
- Linear Algebra (for undergraduates), teaching assistant, Tsinghua University. 2016.9-2017.1
- Calculus (for undergraduates), teaching assistant, Tsinghua University. 2017.2-2017.6
- Advanced Numerical Analysis (for postgraduates), teaching assistant, Tsinghua University. 2017.9-2018.1
- Calculus (for undergraduates), teaching assistant, Tsinghua University. 2018.2-2018.6
- Numerical Analysis (for postgraduates), teaching assistant, Tsinghua University. 2018.9-2019.1

Supervision of students

Master students

- Jiayue Ma, The University of Melbourne, 2023 – 2025 (Co-supervised with Dr. Hailong Guo)

Supervision of students (continued)

- Bohan Sun, The University of Melbourne, 2024 – now (Co-supervised with Dr. Wei Huang)

Skills

Programming
- Advanced: MATLAB, Python
  Basic: C/C++, Linux, JAX

Language
- English, Chinese